TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2015
Higher 2

CANDIDATE NAME	

CENTRE NUMBER	S					INDEX NUMBER					

BIOLOGY 9648/03

Applications Paper and Planning Question **Tuesday, 15 September 2015**

Paper 3 **2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show any working or if you do not use appropriate units.

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/16
2	/10
3	/14
4	/12
5	/20
TOTAL	/72

This document consists of **17** printed pages and **1** blank page.

[Turn over

Answer **all** questions.

1 All commonly cultivated varieties of potatoes are vulnerable to blight, which is caused by a pathogen *Phytophthora infestans*. *P. infestans* kills any plant it infects.

Scientists discovered a blight resistance *Rpi* gene in a species of wild Mexican potato. This gene codes for a plant defence protein that is only expressed by the plant cell when *P. infestans* is present, and it triggers cell death at the site of infection to restrict the growth of *P. infestans*. Infected leaf cells then die, saving the rest of the leaf from infection.

The blight resistance gene can be isolated from the cDNA library.

(a) Describe how this cDNA library can be created using plasmids.

[5]

The recombinant plasmids in the cDNA library were created using restriction enzyme *EcoRI* which has a restriction site on the plasmid pGEM-T.

Fig. 1.1 shows the genetic map of the plasmid pGEM-T.

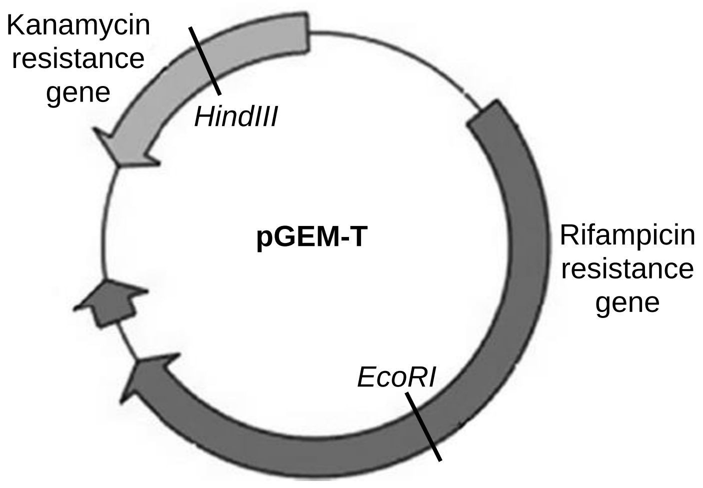

Fig. 1.1

(b) **(i)** Explain how restriction enzymes recognize specific nucleotide sequences.

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____ [2]

(ii) Explain why the bacterial cells carrying the blight resistance gene can be identified using different selection media.

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____ [3]

The transformed bacteria with the blight resistance gene were used to infect plants.

The next generation of transgenic plant can be propagated using explants obtained from these transgenic plants. The plants produced by this process are clones.

(c) **(i)** Outline the technique of cloning new batches of transgenic plants from their explants.

[4]

(ii) Explain why these plants are described as clones.

[2]

[Total: 16]

2 Severe Combined Immunodeficiency (SCID) comprises a group of genetic disorders that are characterized by impaired or non-functional immune system. There are several mutations that can result in SCID, such as mutation to the gene coding for adenosine deaminase (ADA) or the gene coding for the gamma chain of interleukin 2 receptor.

(a) Describe the effects of the mutated ADA gene on a SCID patient.

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____ [2]

One of the ways for long-term treatment of ADA-deficient SCID is to use a combination of viral gene therapy and stem cell therapy to bring the immune system back to normal in the patient. However, immune rejection often occurs when donor stem cells are used.

(b) **(i)** Suggest a suitable virus that can be used as a vector for gene therapy for long-term treatment of SCID.

____ [1]

(ii) Explain briefly how a combination of gene therapy and stem cell therapy can lead to long-term treatment of SCID.

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____ [3]

One of the major problems regarding the use of viruses in gene therapy is the non-specific insertion of normal genes into host cell genome. Recently, scientists have discovered a way to insert genes into specific sites in the genome, without the use of viruses. This technique is known as Clustered, Regularly Interspaced, Short Palindromic Repeat (CRISPR).

The specificity of the insertion is achieved by a short RNA sequence, which is complexed to the Cas9 cutting protein as shown in Fig. 2.1. The Cas9 cutting protein works like an endonuclease.

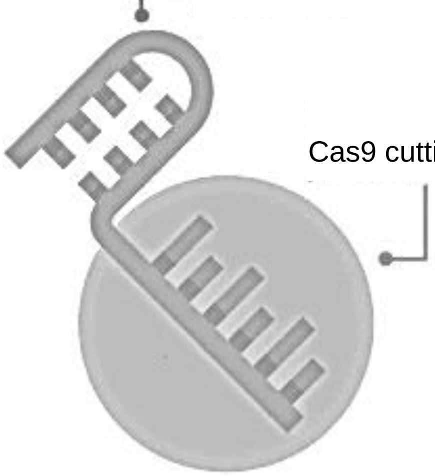

Fig. 2.1

The RNA sequence can be modified by scientists to target specific sites in the genome, so that the normal gene can be introduced into those sites. Fig. 2.2 shows how this can be achieved.

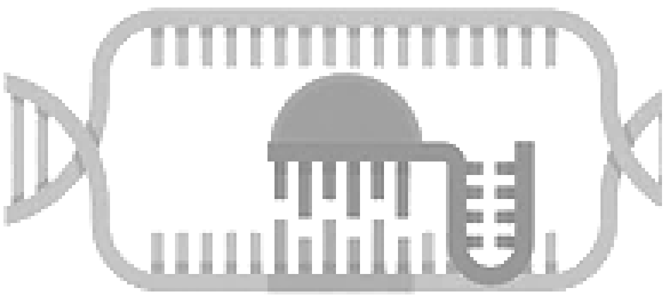

Fig. 2.2

(c) With reference to Fig. 2.1 and 2.2, suggest how the complex achieves specific insertion of a normal gene into the host cell genome.

[2]

The use of viral or CRISPR-mediated gene therapy has raised many ethical issues and concerns.

(d) Discuss **one** ethical issue surrounding somatic cell gene therapy.

[2]

[Total: 10]

3 Golden Rice™ is a genetically modified form of rice that produces relatively large amounts of β-carotene in the endosperm. β-carotene is metabolised in the human body to produce vitamin A.

(a) Explain why rice has been genetically modified to produce β-carotene.

[2]

(b) Suggest **one** economical advantage of producing golden rice.

[1]

Although golden rice has been touted as beneficial, there is much controversy throughout the world regarding the use of genetically modified (GM) crops.

(c) State **three** concerns about the use of GM crops.

[3]

Golden Rice™ was developed using genes from two different organisms, *Psy* (phytoene synthase) gene from daffodil plants and *Crt1* gene from the soil bacterium *Erwinia uredovora*. The transgene, which includes an endosperm-specific promoter, *Psy* cDNA and *Crt1* cDNA, was introduced into the rice endosperm cells.

(d) Suggest why cDNA is used instead of *Psy* and *Crt1* genes.

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____ [2]

The sequence of nucleotides of the *Psy* cDNA can be determined by a method involving PCR followed by gel electrophoresis. In this method, four sequencing reactions are carried out in separate tubes at the same time.

Each tube contains:
- DNA polymerase
- radioactively-labelled primers
- template DNA which is single-stranded
- a mixture of DNA nucleotides (dTTP, dATP, dGTP, dCTP)
- one of four types of modified nucleotides (ddTTP, ddATP, ddGTP, ddCTP)

The modified nucleotide is also known as a replication-stopping nucleotide. This is because the incorporation of a modified nucleotide during the synthesis of the new strand stops the replication process.

Fig. 3.1 shows the positions of the bands produced from the tubes containing ddTTP, ddATP, ddGTP and ddCTP. Only a segment of the *psy* cDNA is shown.

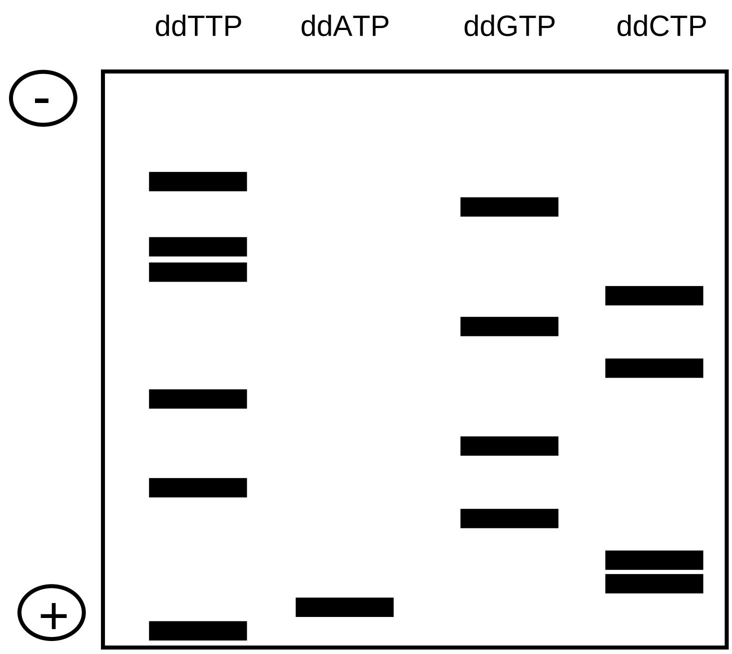

Fig. 3.1

(e) With reference to Fig. 3.1, deduce the sequence of the template strand.

____ [1]

Based on the sequence of the template strand, a 15 base pair long radioactive probe is designed to locate the *Psy* cDNA.

(f) State the sequence of the radioactive probe that could be used to locate the cDNA.

____ [1]

Crt1 cDNA was also isolated using the same method and it was used to synthesize the transgene together with *Psy* cDNA. The transgene was inserted into a plasmid before it is introduced into the rice endosperm cell.

(g) Other than the use of bacteria, state **one** method that can be used to introduce the recombinant plasmid into rice endosperm cells.

____ [1]

Investigations were carried out to determine if *Psy* genes taken from different species would enable rice endosperm to produce greater quantities of β-carotene as compared to original source of *Psy* gene. This was carried out by:

1. inserting *Psy* cDNA from maize, tomatoes, peppers and daffodils into different plasmids, together with the *Crt1* cDNA and an endosperm-specific promoter.

2. introducing the four types of recombinant plasmids into different cultures of rice endosperm cells.

3. measuring the quantity of β-carotene produced by these rice endosperm cells.

The results are shown in Table 3.1.

Table 3.1

Source of *Psy* cDNA gene	**Total β-carotene of rice endosperm cells / arbitrary units**
Maize	14
Pepper	4
Tomato	6
Daffodil	1

(g) With reference to Table 3.1 and the information provided, discuss whether the hypothesis that the production of β-carotene in golden rice is limited by the source of the *Psy* gene is correct.

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____ [3]

[Total: 14]

4 Planning Question

Different plant species have different compositions of pigments, which results in different Photosynthetically-Active Radiation (PAR).

The amount of light absorbed at different wavelengths by a solution/suspension can be determined using a spectrophotometer (Fig. 4.1a), by filling a cuvette with the solution/suspension and placing it in the sample holder (Fig. 4.1b). The wavelength (400 – 700nm) can be adjusted on the spectrophotometer (using the Wavelength Control knob) and the amount of light absorbed is then measured in terms of "Absorbance/Au".

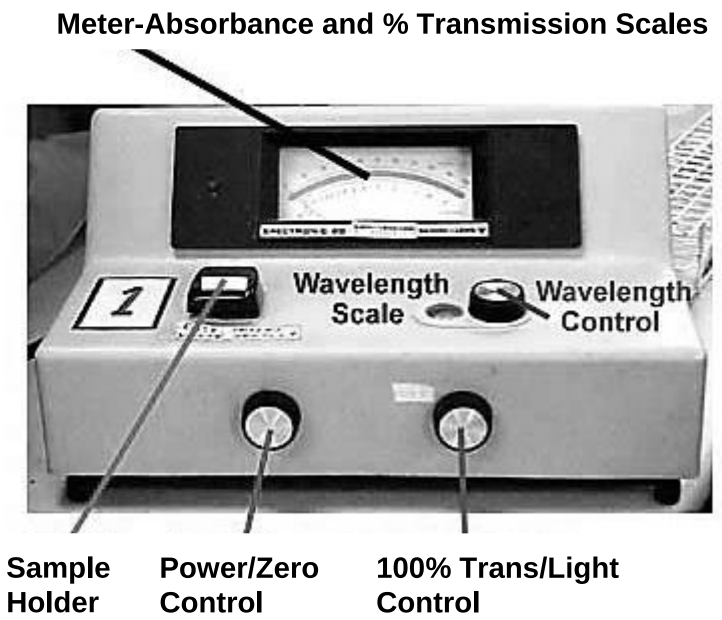

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Fig. 4.1a

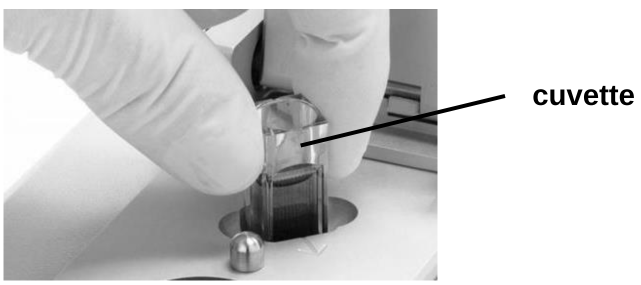

Fig. 4.1b

Using your knowledge and the information provided, design an experiment to determine the absorption spectra of two plant species, **P** and **Q**.

You must use:

- Plant species **P**,
- Plant species **Q**,
- Pigment extraction buffer [✖ IRRITANT],
- Mortar and pestle,
- Fine sand,
- Filter funnel,
- Muslin cloth

You may select from the following apparatus and chemicals:

- Normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 12]

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Explain the advantages and limitations of polymerase chain reaction (PCR). [8]

(b) Explain what is meant by restriction fragment length polymorphism (RFLP) and how it can be detected. [6]

(c) Discuss the aims of the human genome project. [6]

[Total: 20]

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